

HAT-P-36b

R-1715

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-36_200221 · created 2026-08-02T16:00:01+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458900.7986 +/- 0.0033 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.127 +/- 0.044
Transit depth (Rp/Rs)^2	1.61 +/- 1.13 [%]
Orbital Inclination (inc)	87.53 +/- 2.79
Airmass coefficient 1 (a1)	0.978 +/- 0.047
Airmass coefficient 2 (a2)	0.025 +/- 0.08
Scatter in the residuals of the lightcurve fit is	4.81 %
Best Comparison Star	#1 - [137, 119]
Optimal Aperture	2.3
Optimal Annulus	6.24
Transit Duration (day)	0.0948 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.127 $\pm$ 0.044 vs 0.126 (deviation 0.02 σ)
Duration fit vs published	0.0948 d vs 0.0925 d (deviation 0.38 σ)
Mid-time O-C	-8.4 min
Depth SNR	2.9
Residual scatter	4.81 %

Light curve

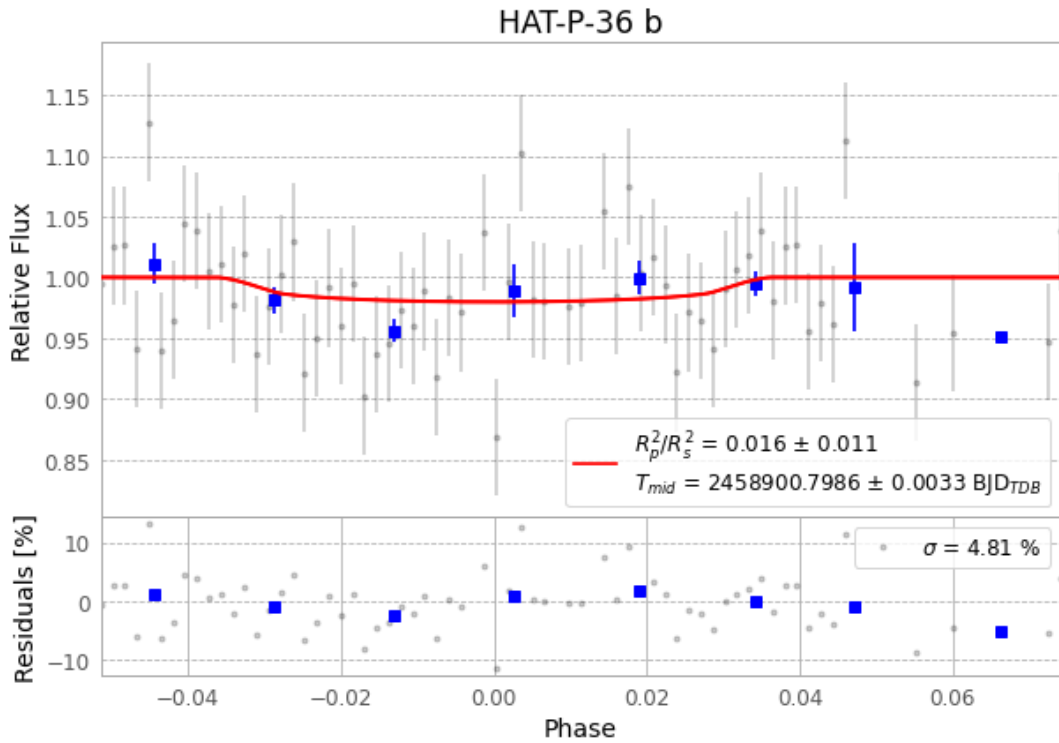


Figure 1 — FinalLightCurve_HAT-P-36 b_2020-02-20.png (SHA-256 1c28453791bbfa47...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [408, 187], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d2a878af2167aa3e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

81 FITS frames (53 MB), HATP-36200221052412.FITS → HATP-36200221092412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-36-200221.log (7c7ab0395502...), FinalLightCurve_HAT-P-36 b_2020-02-20.png (1c28453791bb...), FinalLightCurve_HAT-P-36 b_2020-02-20.csv (1814e92d6373...)

Compute resources

Wall time 240.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 16:00Z.